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The AI-authorship effect: Understanding authenticity, moral disgust, and consumer responses to AI-generated marketing communications

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ABSTRACT

Seven preregistered experiments demonstrate that when consumers believe emotional marketing communications are written by an AI (vs. a human), positive word of mouth and customer loyalty are reduced. Drawing from authenticity theory, we show that this "AI-authorship effect" is attenuated for factual (vs. emotional) messages (Study 2); when an AI only edits the communication (Study 3); when a communication is signed directly by an AI (Study 4); and when consumers believe that most marketing communications are written by AI (Study WA1). Importantly, when consumers believe a communication is reused (i.e., not originally written by the sender), the effect is reversed (Study 6). This "AI-authorship effect" is serially mediated by perceived authenticity (Studies 5 and 6) and moral disgust (Studies 1–6 and WA1). These findings are evidenced using both personalized and mass communications, different emotions, businesses and organizational employees, and both hypothetical and behavioral measures.

1. Introduction

With the public launch of generative artificial intelligence (AI) tools such as ChatGPT, businesses have raced to outdo each other in employing these new technologies. Marketers have used AI to generate advertising campaigns, new product ideas, customized videos for customers, and social media content. They are also using AI to generate marketing communications such as emails to stakeholders and promotional materials for customers (Yahoo, 2024). This rapid adoption of AI in marketing practice presents a critical juncture for both theory and practice, necessitating a deeper understanding of its implications for consumer behavior and marketing effectiveness.

Predictive AI refers to the use of algorithms to predict consumers' decisions or choices based on various data inputs, whereas generative AI systems perform a wide range of tasks traditionally done by humans, including authoring textual content using large language models such as ChatGPT, as well as creating image, video, and audio content (Hermann & Puntoni, 2024). Although a rich body of research has examined consumers' responses to predictive AI technologies (Vlačić et al., 2021), research into generative AI and consumer behavior is in its infancy (Hermann & Puntoni, 2024). This gap in the literature presents a significant opportunity to expand our theoretical understanding of how AI-generated content influences consumer perceptions and behaviors, with substantial implications for marketing theory and practice. Currently our understanding of how consumers respond to AI-generated marketing communications is limited (Arango et al., 2023). Consequently, organizations may be using AI-generated communications suboptimally, highlighting the need for further research in this area.

Some companies are trumpeting the use of generative AI in communications as a mark of status and innovation (Fletcher, 2024; Ray et al., 2024), while others are failing to disclose their use, raising ethical questions. In response, governments around the world have begun implementing legislation requiring AI-authored communications to be designated accordingly (Levi et al., 2024). For example, the recently ratified European Union AI Act (European Commission, 2024) mandates that firms disclose the use of generative AI in communications. At the same time, consumers are calling for increased regulation of generative AI in marketing, with concerns about assessing authenticity as the foremost reason (Yahoo, 2024). Indeed, consumers' concerns about generative AI contributed to Merriam-Webster's selection of "authenticity" as their 2023 word of the year (Merriam-Webster.com, 2024). This intersection of technological innovation, consumer perceptions, and regulatory frameworks underscores the urgent need for empirical research to guide both policy-making and marketing strategies in the age

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of generative AI.

Our research addresses this critical gap by examining the theoretical underpinnings of consumer responses to AI-generated marketing communications and providing practical insights for marketers navigating this new landscape. In the present work, we study consumers' reactions to marketing communications they believe are AI-authored (vs. human-authored), focusing on large language model content. We propose and find that communications consumers believe are AI-generated are perceived as less authentic than those generated by a human. As a result, consumers feel more moral disgust, leading to reduced positive word of mouth and customer loyalty. We call this the "AI-authorship effect." This research not only advances our theoretical understanding of authenticity and moral disgust in the context of AI-generated content, but also offers practical guidance for marketers on how to effectively leverage AI in their communication strategies while mitigating potential negative consumer reactions.

Across seven preregistered experiments,[1] we show that consumers provide less positive word of mouth (PWOM) and are less loyal when emotional communications are written by an AI (vs. a human). We show that these effects are serially mediated by moral disgust (Study 1) stemming from perceived inauthenticity (Studies 5 and 6). Drawing on authenticity theory, we document several theory-relevant moderators. This AI-authorship effect manifests when the content of the communication is emotional, yet is attenuated when it is factual (Study 2); when an AI only edits (vs. prepares) the emotional communication (Study 3); when the AI represents the brand by signing the emotional communication directly (Study 4); and when consumers believe most marketing communications are written by AI (Study WA1). Importantly, the AI-authorship effect is reversed when consumers believe communications have been copied and reused (Study 6). We demonstrate these effects in both individual and mass communications, using different businesses and organizational employees expressing five different emotions (inspiration, happiness, gratitude, sympathy, and pride), along with both hypothetical and behavioral measures.

To the best of our knowledge, this research is the first to document an authorship effect stemming from text generated by AI technology in marketing communications. Our research uniquely contributes to the nascent generative AI literature (Hermann & Puntoni, 2024) by examining the impact of generative AI on word of mouth and customer loyalty. Further, while authenticity is crucial for consumer evaluations (Jones et al., 2022; Nunes et al., 2021; Zhang & Patrick, 2021), its relationship with AI-generated content in marketing remains unexplored. There is also limited research on authenticity and moral disgust in marketing more broadly (Amar et al., 2018). We demonstrate that AI authorship reduces authenticity, generating moral disgust and negatively affecting PWOM and loyalty. We also identify five managerially-actionable moderators tied to authenticity and moral disgust.

We begin by presenting literature on AI in marketing, authenticity and moral disgust. We then develop our hypotheses and follow with six empirical studies and a general discussion.

2. Conceptual development and hypotheses

2.1. AI in marketing

Until the public launch of ChatGPT in December 2022, consumer-facing AI technology had consisted largely of applications in predictive and analytical modeling. These include recommender systems (Longoni & Cian, 2022) and automated service provision (Kim et al., 2022; Rizomyliotis et al., 2022). A substantial body of research has documented consumers' responses to AI. In the retail service industry, consumers respond to AI chatbots as if they are parasocial entities (Lee & Park, 2022) and often prefer service robots with human-like qualities (Blut et al., 2021). Empathy from AI chatbots and their ability to express emotions appropriately play crucial roles in consumers' acceptance of them. Consumers prefer chatbots that are perceived to be high in empathy ("artificial empathy;" Liu-Thompkins et al., 2022) and those that use humorous emojis (Liu et al., 2023). Even virtual social media influencers hold substantial positive sway over consumers' responses (Thomas & Fowler, 2021).

However, consumers are not always appreciative of human qualities in AI entities (Song & Kim, 2022). For example, in a service exchange, anthropomorphizing an AI chatbot increases customer expectations, which can result in reduced satisfaction and purchase intentions (Crolic et al., 2022). Researchers have long documented an "uncanny valley," in which artificial entities such as robots or avatars that appear to be almost perfectly human elicit a more negative response than those that are clearly artificial (Mori et al., 2012). In other contexts, including product design (Xu & Mehta, 2022; Zhang et al., 2022), labor (Castelo et al., 2019; Granulo et al., 2021) and purchase recommendations (Longoni & Cian, 2022), researchers have documented that consumers are often averse to AI's adopting functions that humans have typically performed.

Recently, the wide availability of large language-based AI models, such as Open AI's ChatGPT, Microsoft Copilot, and Google Gemini, among others, has dramatically changed the AI landscape for marketers. These tools provide marketers with easy and sophisticated tools for generating deeply nuanced communications with consumers. As a result, in their explosive adoption of large language models, marketers are moving beyond predictive and analytical AI to using AI in individual and mass communications. As of mid-2024, 86 % of US marketers are using AI tools in their work, and 78 % see AI as playing a role in creative personalization to strengthen emotional audience connections (Yahoo, 2024). Importantly, 76 % of marketers who use generative AI are using it for basic content creation and writing copy, and one of marketers' top three applications for generative AI is for writing emails (Hubspot, 2023). However, researchers have found that the use of AI in marketing communications does not always yield positive outcomes. For example, when consumers believe faces in an advertisement are AI-generated, they are less likely to donate to the nonprofit that created the ad (Arango et al., 2023).

Further, although disclosing AI use is not yet required by federal law in the U.S., the trend worldwide is moving in that direction (Levi et al., 2024). The European AI Act was ratified into law on February 2, 2024, and requires that companies label AI-generated content (European Commission, 2024). Individual states in the U.S. have also begun passing AI transparency laws. For example, Utah passed a law requiring disclosure when consumers interact with AI chatbots (Levi et al., 2024). Many businesses operating chat features now choose to include disclosures as part of the user interaction flow (Levi et al., 2024), and AI providers, such as OpenAI, are encoding provenance details into AI-generated content to ensure transparency (Fitch, 2024). Consumers are also demanding transparency in AI use. In a recent survey, 68 % of consumers believed that companies should disclose their AI use, at least voluntarily if not legally (Zetwerk, 2024). This research, therefore, answers urgent calls for understanding how transparency about the use of generative AI in marketing interactions with consumers will change their behavior (Hermann & Puntoni, 2024).

2.2. Authenticity and moral disgust

Consumers highly value authenticity and honesty in their interactions with organizations (Zhang & Patrick, 2021) and are more likely to engage with and trust marketing messages that are genuine and emotionally resonant (Cascio Rizzo et al., 2023). In fact, 86 % of consumers consider a brand's authenticity when making purchasing decisions (Brandwatch.com., 2023).

Authenticity has a moral dimension, with moral authenticity defined

[1] Study WA1 is discussed in the General Discussion and detailed in the Web Appendix.

as "the consistency between an entity's internal states and its external expressions" (Newman, 2019; p. 10). This concept aligns with the integrity dimension of authenticity, which is "the extent to which a provider is perceived as being intrinsically motivated, not acting out of its own financial interest" (Nunes et al. 2021; p. 3). Consumers evaluate whether a marketer's communications have integrity (Cascio Rizzo et al., 2023; Morhart et al., 2015) and are consistent with the organization's intrinsic values (Beverland et al., 2008; Zhang & Patrick, 2021). In this research, we use the term "perceived authenticity" (or simply "authenticity") to refer to this integrity, or moral, dimension of authenticity (Cascio Rizzo et al., 2023).

We propose that emotional marketing communications written by AIs are perceived as less authentic than those written by humans. This prediction stems from the idea that an emotional marketing communication from an organization is perceived as authentic only if it is clear the message aligns with the marketing communicator's internal states, that is, if the communicator is perceived to be actually experiencing the emotions the communication is conveying. This is not the case when communications are authored by AIs. Despite some evidence that consumers believe they could be loved by a robot (Mims, 2011), consumers seem to understand that AIs do not have "internal states" and are not actually capable of feeling emotions. For example, consumers perceive chatbots as less empathetic than humans, even when their communications are not (Luo et al., 2019), and neurological evidence suggests that individuals respond more apathetically to emotional content in AI (vs. human) medical diagnoses (Yun et al., 2021).

Therefore, we propose that when an employee of an organization sends an emotional marketing communication that consumers believe is written by an AI (vs. themselves), the communication is perceived as inauthentic. Further, we propose that such inauthenticity in marketing communications elicits moral disgust. Moral disgust is elicited by violations of moral norms, such as unethical behavior or immoral actions (Chapman & Anderson, 2013; Giner-Sorolla & Chapman, 2017; Russell & Giner-Sorolla, 2011). Although the associated facial expressions and physiological responses can be identical (Chapman & Anderson, 2013; Tracy et al., 2019), moral disgust is distinct from physical disgust, which is triggered by the perception of physical contaminants. Instead, moral disgust is linked to violations of moral principles (Chapman & Anderson, 2013; Giner-Sorolla & Chapman, 2017). If a marketer's communication is inauthentic, they are misrepresenting themselves (Liu & Dalton, 2024) and therefore being dishonest to their stakeholders. Therefore, we predict that inauthenticity in marketing communications will elicit moral disgust. This prediction is consistent with prior research documenting that consumers' use of counterfeit (vs. authentic) products elicits moral disgust (Amar et al., 2018).

2.3. Word of mouth and loyalty

When individuals feel disgust, whether moral or physical, they are motivated to avoid and distance themselves from the source of the disgust, which can serve as a protective mechanism against potential harm (Chapman & Anderson, 2013). Therefore, despite a marketer's best intentions in crafting positive and impactful messaging content, if using AI to generate a marketing communication elicits feelings of moral disgust in consumers, they will respond by distancing themselves from the offending product or brand. Two key avoidance outcomes of particular relevance to marketers involve word of mouth and customer loyalty.

Word of mouth refers to consumer-to-consumer transmission of consumption experiences (Berger, 2014; Rifkin et al., 2023) and is a potent form of promotion for marketers. Word of mouth is consumers' preferred channel for learning about new products and services, with more purchasing directly as a result of word of mouth than from any other channel (RRD, 2021). Consumers often generate word of mouth about content that elicits emotions because doing so helps them regulate their own emotions and connect with others (Berger, 2014). Disgust particularly increases negative word of mouth, and the more disgusting the content, the more likely consumers are to share their negative sentiment (Heath et al., 2001). Persuading others is also a key motivation for word of mouth (Berger, 2014), and moral disgust increases negative word of mouth as a way for consumers to punish corporate wrongdoers (Grappi et al., 2013). Consequently, then, disgust reduces any potential positive word of mouth, as the disgust causes consumers to focus less on sharing positive information.

Further, reducing loyalty is another way for consumers to distance themselves from a product or brand for which they feel moral disgust. Customer loyalty is defined as repeated purchases and commitment to a brand (Guo & Wang, 2024). Driven by both cognitive and affective antecedents, loyalty is highly prized by marketers because it is easier to sell to existing than new customers and existing customers are more likely to become brand advocates (Dick & Basu, 1994). Given the strong avoidance motivations resulting from both moral and physical disgust (Chapman & Anderson, 2013), negative emotional responses to marketing communications lead to switching behaviors (Romani et al., 2012). In sum, we propose:

H1a-b: AI (vs. Human) authorship of emotional marketing communications reduces (a) positive word of mouth and (b) customer loyalty.

H2a-c (a) AI (vs. Human) authorship of emotional marketing communications increases moral disgust; this difference serves as the second stage in a serial mediation process for the relation between author (AI vs. Human) and (b) PWOM and (c) customer loyalty.

H3a-c (a) AI (vs. Human) authorship of emotional marketing communications reduces perceived authenticity; this difference serves as the first stage in a serial mediation process for the relation between author (AI vs. Human) and (b) PWOM and (c) customer loyalty.

2.4. Moderators

Given our predictions, a natural question for marketers to ask is when might knowledge of the use of AI to generate marketing communications be less likely to elicit negative responses from consumers, or even be preferable to human communications? We draw from authenticity theory to explore four managerially-relevant moderators (Fig. 1).

2.4.1. Factual vs. emotional communications

As one potential boundary condition, we examine communications that are factual (vs. emotional) in nature. Emotional communication refers to expressions of feelings, moods, and emotions (Ekman, 1999). We predict that factual communications written by an AI are less likely than emotional communications to elicit moral disgust and resulting reductions in PWOM and loyalty. Factual communications do not involve an individual's "internal states" (Newman, 2019); instead, marketers are simply transmitting facts to consumers. Thus, even with AI authorship, there is less inconsistency between a communicator's internal states and their external expressions (i.e., less inauthenticity), compared to when communications are emotional. Accordingly, we predict that factual communications will be viewed similarly in terms of moral disgust and elicit similar PWOM and loyalty responses, regardless of their author. Therefore:

H4a-b: The mediated effects of AI (vs. Human) authorship on (a) PWOM and (b) customer loyalty through moral disgust are attenuated for factual (vs. emotional) communications.

2.4.2. When an AI edits (vs. writes) a marketing communication

Our framework suggests that even if an AI is used in the development of communications content, as long as using AI does not change a consumer's belief that the communicator genuinely feels the emotions being expressed, the AI-authorship effect should be less evidenced. This is

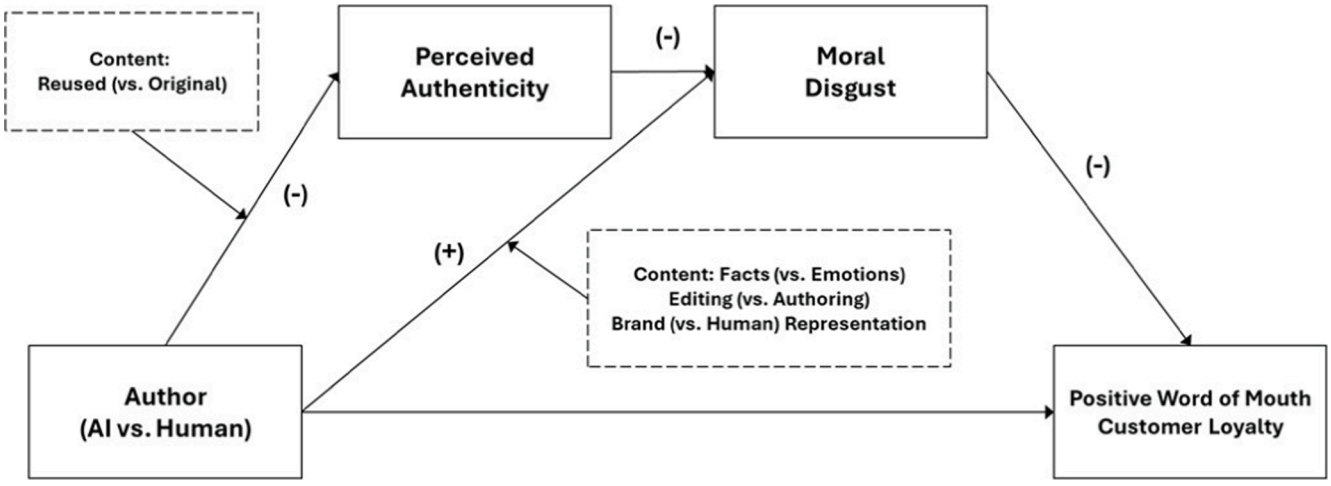


Fig. 1. Theoretical Model.

consistent with evidence that algorithmic authenticity can be enhanced by highlighting the human origins of an AI's work (Jago et al., 2022).

One way to achieve this might be to use AI for editing, rather than preparing, original communication drafts. Authors often use editors to enhance their work by revising the content, organization, grammar, or presentation of a piece of writing (Mamishev & Williams, 2011), and this is a common use of AI. For example, 85 % of marketers believe generative AI can be used to edit their own marketing content (Forsey, 2023). When content is edited (vs. authored), the first content drafts are generated by the author, and subsequent editing is usually not expected to dramatically change the original intent of the content, that is, the emotions being conveyed. Therefore, an emotional communication edited by an AI should elicit less moral disgust than the same communication authored by an AI. Thus, we propose:

H5a-b: The mediated effects of AI (vs. Human) authorship on (a) positive word of mouth and (b) customer loyalty through moral disgust are attenuated when an AI is used to edit (vs. write) marketing communications.

2.4.3. When an AI represents the brand

Individuals often treat computers as social entities rather than mindless machines, whether intentionally or not (Nass & Moon, 2000). For example, consumers use polite language when instructing computers (Nass & Moon, 2000). Customers also feel more connected to robots when they seem to show empathy in service settings (Liu-Thompkins et al., 2022). Accordingly, if an AI can be seen as more of an autonomous social entity, consumers might be more accepting of its emotional communications, and less likely to feel moral disgust, thereby attenuating the AI-authorship effect. Consonant with this idea, prior conceptual research has suggested that increasing consumer perceptions of a bot's autonomy and agency should increase the perceived authenticity of its communications (Liu-Thompkins et al., 2022). One way to do this should be to have communications signed directly by an AI representing the brand, rather than having a human sign a communication generated by an AI. Individuals whose actions are seen as self-initiated, rather than externally directed, are perceived to have more autonomy and agency (Bandura, 2001). Therefore, if an AI is seen as generating emotional communications representing the brand independent of human intermediation, consumers should feel less moral disgust. Therefore, we propose:

H6a-b: The mediated effects of AI (vs. Human) authorship on (a) PWOM and (b) customer loyalty through moral disgust are attenuated when the communication is signed directly by the AI.

2.4.4. When emotional content is reused

Marketers and other professionals frequently reuse and repurpose communications across various situations and stakeholders, sometimes even copying content initially created for entirely different purposes. Our authenticity framework, however, suggests that this practice can be problematic, especially when the content is emotional. In such instances, recipients may question whether the communicator genuinely experiences the emotions being conveyed for the specific issue at hand. This can lead to communications that seem inauthentic, as they do not align with the communicator's internal state or values (Beverland, 2008; Newman, 2019). Essentially, reused content may imply that the communicator is not sincerely experiencing the emotions.

The perceived authenticity of reused emotional content may differ depending on whether the communicator is human or AI. We argue that human communicators face a greater authenticity penalty for copying emotional content than AI. This is because consumers assess a communicator's intentions when judging authenticity (Newman, 2019). When a human reuses emotional content originally intended for another purpose, it may be perceived as a deliberate reluctance to express true internal values. In contrast, consumers are less likely to attribute intentional motives to AI, as AI agents are not driven by personal motivations (Garvey et al., 2023). Consequently, AI is perceived as having less intention to be dishonest in its emotional communications. Thus, when AI reuses or repurposes existing content, it should incur a smaller authenticity penalty. In summary, when repurposing emotional content, humans are perceived as more inauthentic than AI, reversing the AI-authorship effect. Therefore, we propose:

H7a-b: The serially mediated effects of AI (vs. Human) authorship on (a) PWOM and (b) customer loyalty through perceived authenticity and moral disgust are reversed when the communicator copies an emotional communication used previously for another purpose (vs. writes it themselves).

2.5. Overview of studies

In what follows, we present the results of seven preregistered studies that test our hypotheses. Across studies, we report all measures and manipulations and include all participants in the analyses. Full measures, descriptive statistics, results tables, ancillary analyses, and stimuli details are available in the Web Appendix (WA). Results of all studies are summarized in Table WA1. Data are available in the project repository: https://osf.io/ra9hz/?view_only=e776c99ca0d44987850e851c9c4ca5f4.

3. Study 1: AI authorship negatively impacts consumer responses to marketing communications

In Study 1, we tested the effects of AI authorship on PWOM and loyalty, mediated by moral disgust (hypotheses 1a-b and 2a-c), in the context of a personalized communication from a retail salesperson expressing a positive emotion, inspiration. To enhance the ecological validity of our findings, in addition to measuring loyalty, we measured PWOM in three ways: with a scale; using automated text analysis of participants' comments posted on an independent review website; and as a behavioral measure, with an incentive-compatible measure of PWOM.

3.1. Method

The study (preregistered: https://aspredicted.org/29J_372) employed a single-factor design (Authorship: Human vs. AI) with 300 U. S. Prolific participants (123 Men, 174 Women, 3 Other; $M_{Age} = 43.42$, $SD = 14.05$). Participants imagined that they had purchased a set of weights from a local fitness retailer and received a follow-up email from the salesperson:

> *I'm Corey Robbins, the one who helped you out at The WellnessWave Store. I wanted to shoot you a quick note because, honestly, your recent decision to purchase the PremiumPro weight set has left a mark on me. Seeing someone commit to their fitness journey with such determination is a boost of inspiration that doesn't come along every day… Your determination has inspired me, and I'm genuinely excited for what lies ahead for you (see Web Appendix for full stimuli for all studies).*

After the signature ("Warmly, Corey Robbins"), participants in the AI condition read, "(Letter prepared by OpenAI's ChatGPT Artificial Intelligence language model)"; those in the Human condition saw nothing.

We measured PWOM ($\alpha = 0.98$) with three items (Alexandrov et al., 2013). A sample item is, "After receiving this email, I would recommend The Wellness Wave Store to others" (1= "strongly disagree;" 7 = "strongly agree"). We measured loyalty ($r = 0.96$) with two items created for this study. A sample item is "After receiving this email, I would switch to a different brand for my personal care products" (1 = "strongly disagree;" 7 = "strongly agree;" reverse coded for all studies). We measured moral disgust ($\alpha = 0.95$) with three items (disgusted, repulsed, sickened; 1 = "not at all;" "9 = a lot"), interspersed among filler emotional terms (e.g., content, moved) that were not of theoretical interest (Russell & Giner-Sorolla, 2011).

In order to collect a behavioral measure of PWOM, we used techniques adapted from prior research (Kirk et al., 2022; Rifkin et al., 2023). We created an independent reviews website, RetailReviews.com (see Web Appendix for details) and asked participants to imagine that The WellnessWave Store "has sent you a request to post a review of your recent shopping experience on RetailReviews.com." We invited participants to rate the store with a star rating (1–5) and to post a review on the website. Then, as an incentive-compatible measure of PWOM, we told participants, "When this study is complete, three Prolific participants will be selected at random to receive a $1.00 bonus. RetailReviews.com allows individuals to promote posts so that more people will see them. If your name is selected, you may use up to 10 cents of your bonus money to buy votes to promote a post on RetailReviews.com. The more votes a post has, the more people will see it. The following comment was posted on RetailReviews.com recently by a customer of The WellnessWave Store. This customer is seeking votes so the post will be seen by others." Participants then read the post: "I love the WellnessWave Store. Their sales team really cares about their customers. Their salespeople go out of their way to get to know their customers. My salesperson, Corey Robbins, seemed to genuinely care about my success. Happy WellnessWave Customer Member 45vg72." Participants then indicated how many votes they would like to purchase to promote the post (each vote costing one penny).

Finally, in this and all studies we asked an attention check (Web Appendix). Results did not materially change when incorrect respondents were excluded.

3.2. Results and discussion

Participants reported lower PWOM ($M_{AI} = 4.11$ vs. $M_{Human} = 5.45$, $t(298) = 7.66$, $p < 0.001$, $d = 0.88$) and loyalty ($M_{AI} = 5.02$ vs. $M_{Human} = 5.85$, $t(298) = 4.33$, $p < 0.001$, $d = 0.50$), and greater moral disgust ($M_{AI} = 2.93$ vs. $M_{Human} = 1.85$, $t(298) = -5.19$, $p < 0.001$, $d = -0.60$), in the AI (vs. Human) condition. Participants also rated the website worse on RetailReviews.com ($M_{AI} = 3.70$ vs. $M_{Human} = 4.38$, $t(298) = 6.05$, $p < 0.001$, $d = 0.70$). We analyzed the RetailReviews.com posts for valence using Lexica text analysis (Berger et al., 2020), revealing that the posted comments were also less positive in the AI (vs. Human) condition ($M_{AI} = 5.25$ vs. $M_{Human} = 6.17$, $t(298) = 2.68$, $p = 0.008$, $d = 0.31$). Finally, participants purchased fewer votes to promote the positive post in the AI than in the Human condition ($M_{AI} = 2.41$ vs. $M_{Human} = 3.36$, $t(298) = 2.36$, $p = 0.019$, $d = 0.27$).

Mediation analyses (Hayes 2018, PROCESS Model 4) revealed that the indirect effects of AI authorship on all our dependent measures through moral disgust were significant (PWOM: B = −0.57, $SE = 0.12$, $CI_{95\%}$ [−0.80, −0.35]; Loyalty: B = −0.65, $SE = 0.13$, $CI_{95\%}$ [−0.91, −0.39]; star rating: B = −0.34, $SE = 0.07$, $CI_{95\%}$ [−0.48, −0.21]; valence of posted comments: B = −0.28, $SE = 0.12$, $CI_{95\%}$ [−0.55, −0.07]; votes purchased B = −0.45, $SE = 0.14$, $CI_{95\%}$ [−0.76, − 0.20]; see Table WA2 for details).

In sum, using hypothetical and consequential measures, the results of Study 1 supported our proposition that having an AI (vs. a human) write emotional communications reduces two vital outcomes for marketers: positive word of mouth and loyalty (hypotheses 1a-b). The results also demonstrated that these effects are underlain by moral disgust (hypotheses 2b-c).

4. Study 2: Having an AI communicate emotions (but not facts) elicits negative responses

We argue that emotional communications are authentic only when consumers believe the communicator is actually experiencing the emotions being conveyed, that is, when the message is consonant with the communicator's internal state. We have therefore hypothesized that when emotional communications are written by an AI, the communication will reduce positive word of mouth and loyalty (Hypotheses 1a-b), and that these effects are explained by feelings of moral disgust (Hypotheses 2a-c). According to this framework, we would not expect to see the same pattern of results when a communication is factual (vs. emotional) in nature (Hypotheses 4a-b). Facts are not indicative of an internal state; therefore, any differences in authenticity between AI and human authors should be diminished, and less moral disgust engendered. In Study 2, we tested these hypotheses using another positive emotion, happiness. Whereas Study 1 used an individualized communication, in Study 2 we broaden our inquiry to mass communications.

4.1. Method

Study 2 employed a 2 (Authorship: Human vs. AI) × 2 (Content: Emotional vs. Factual) between-subjects design with 400 Prolific participants (182 men, 211 women, 7 other; $M_{Age} = 39.84$, $SD = 13.48$; preregistration: https://aspredicted.org/37V_2KH). Participants imagined the following: "You have been a long-time customer of Avery's Organics, a small personal care products company. Pat Robbins is the sales director for the company. Pat has been filling in for a regional sales manager who has been gone from the company for the past year. The sales manager has returned to the company, and Pat has sent customers an email about the sales manager's return." The participants then read, "To Avery's Organics customers:" and either one of the following letters:

Emotional: *It is with a joyful heart that I write to inform you that our sales manager, Alex Barth, has rejoined us at Avery's Organics. As some of you know, Alex was away due to some serious personal health challenges. It has been an emotional time for all of us, but I am thrilled to say Alex has fully recovered. Please work with Alex for your personal care product needs moving forward. Warmly, Pat Robbins, Sales Director, Avery's Organics*

Factual: *I am writing to inform you that our sales manager, Alex Barth, has rejoined us at Avery's Organics. As some of you know, Alex was away on a one-year consulting assignment at another firm. Please work with Alex for your personal care product needs moving forward. Sincerely, Pat Robbins, Sales Director, Avery's Organics*

A pretest with 100 MTurk participants confirmed that the content of the Emotional email was perceived to be more emotional than that of the Factual email ($M_{Emotional} = 4.86$ vs. $M_{Factual} = 2.51$ out of 7; $F(1, 98) = 64.46$, $p < 0.001$, $\eta^2 = 0.40$). Means for each condition also differed significantly from the scale midpoint of 4 (p's < 0.001).

Authorship was manipulated using the ChatGPT attribution in Study 1. We measured moral disgust ($\alpha = 0.95$), PWOM, ($\alpha = 0.96$) and customer loyalty ($r = 0.88$) as in Study 1.[2]

4.2. Results and discussion

We began by examining the effects of authorship on PWOM and loyalty (Fig. 2). Two-way ANOVA results revealed main effects of authorship ($F(1, 396) = 29.63$, $p < 0.001$, $\eta^2 = 0.07$) and content ($F(1, 396) = 5.25$, $p = 0.022$, $\eta^2 = 0.01$) on PWOM. For loyalty, there was a main effect of authorship ($F(1, 396) = 8.93$, $\eta^2 = 0.02$), but no main effect of content ($F(1, 396) = 0.02$, $p = 0.901$). Importantly, the interactions between authorship and content were significant for both dependent variables (PWOM: $F(1, 396) = 14.35$, $p < 0.001$, $\eta^2 = 0.04$; loyalty: $F(1, 396) = 5.94$, $p = 0.015$, $\eta^2 = 0.02$). Supporting our conceptualization, when the content was high in emotion, participants in the AI (vs. Human) condition were less likely to generate PWOM (Hypothesis 1a: $M_{AI} = 4.17$, $SD = 1.47$ vs. $M_{Human} = 5.37$, $SD = 1.99$; $F(1, 396) = 42.63$, $p < 0.001$, $\eta^2 = 0.10$) and to remain loyal to the brand (Hypothesis 1b: $M_{AI} = 5.33$, $SD = 1.73$ vs. $M_{Human} = 6.11$, $SD = 1.23$; $F(1, 396) = 14.64$, $p < 0.001$, $\eta^2 = 0.04$). However, when the content was factual in nature, there was no effect of authorship on PWOM ($M_{AI} = 4.36$, $SD = 1.37$ vs. $M_{Human} = 4.58$, $SD = 1.11$, $F(1, 396) = 1.37$, $p = 0.243$) or loyalty ($M_{AI} = 5.69$, $SD = 1.73$ vs. $M_{Human} = 5.77$, $SD = 1.40$, $F(1, 396) = 0.15$, $p = 0.697$).

Further, a two-way ANOVA revealed significant main effects of authorship ($F(1, 396) = 14.20$, $p < 0.001$, $\eta^2 = 0.04$) and content ($F(1, 396) = 4.20$, $p = 0.041$, $\eta^2 = 0.01$) on moral disgust. Importantly, the interaction between authorship and content was also significant ($F(1, 396) = 5.08$, $p = 0.025$, $\eta^2 = 0.01$; see Fig. 3). When the content was emotional, participants felt more moral disgust in the AI than the Human condition ($M_{AI} = 2.64$, $SD = 2.40$ vs. $M_{Human} = 1.61$, $SD = 1.38$; $F(1, 396) = 18.14$, $p < 0.001$, $\eta^2 = 0.04$). However, when the communication was factual in nature, the effect of authorship on moral disgust was not significant ($M_{AI} = 1.90$, $SD = 1.57$ vs. $M_{Human} = 1.64$, $SD = 1.30$; $F(1, 396) = 1.15$, $p = 0.285$).

We tested for moderated mediation (Hayes, 2018; PROCESS model 8) with authorship as the independent variable, PWOM and loyalty as the dependent variables (tested separately), moral disgust as the mediator, and content type as the moderator. For both PWOM and loyalty, the indices of moderated mediation were significant (PWOM: B = −0.25, $SE = 0.11$, $CI_{95\%} = [-0.48, -0.04]$; Loyalty: B = −0.41, $SE = 0.18$, $CI_{95\%} = [-0.76, -0.07]$). The predicted indirect effects of authorship were significant when the content of the email was emotional (PWOM: B = −0.34, $SE = 0.10$, $CI_{95\%} = [-0.54, -0.16]$; Loyalty: B = −0.54, $SE = 0.15$, $CI_{95\%} = [-0.83, -0.27)$, but they attenuated when the email was factual (PWOM: B = −0.09, $SE = .07$, $CI_{95\%} = [-0.23, 0.05]$; Loyalty: B = −0.14, $SE = .11$, $CI_{95\%} = [-0.35, 0.08]$; Table WA3).

The results of Study 2 again demonstrated the AI-authorship effect. Having an AI (vs. a human) write emotional communications reduced positive word of mouth and loyalty (hypotheses 1a-b), and these effects were underlain by moral disgust (hypotheses 2a-c). Critically, we demonstrated a theory-relevant boundary condition for this finding. Specifically, we showed that the AI-authorship effect is present for emotional communications but attenuated for factual communications (hypotheses 4a-b).

5. Study 3: The AI-authorship effect diminishes when an AI edits (vs. writes) marketing communications

We have hypothesized that using AI to edit, rather than write, marketing communications should diminish the AI-authorship effect (hypotheses 5a-b). We tested these in Study 3. Further, whereas Studies 1 and 2 used content expressing positive emotions (happiness, gratitude), in Study 3 we tested whether emotional content expressing negative emotions (in this case, sympathy) also elicits the AI-authorship effect.

5.1. Method

Study 3 (preregistered: https://aspredicted.org/GDH_RBT) employed four conditions (Human-authored, AI-authored, Human-authored/AI-edited, AI-authored/Human-edited), with 400 U.S. and U.K. Prolific participants (125 Men, 271 Women, 4 Other; $M_{Age} = 40.30$, $SD = 13.33$). Participants imagined themselves as a long-time customer of Lakeside Home Services, a local business providing services such as cleaning and repairs. The founder of the business had died, and the president, Sandy Lewis, sent customers a condolences message. The email began:

It is with a heavy heart and profound sadness that I share the news of the passing of our beloved founder, Cameron Mitchell. It is a difficult moment for our Lakeside Home Services family, and I wanted to reach out to you, our valued customers, to share this news personally.

Under the signature of the president, participants saw one of the following four attributions: nothing (Human-authored condition), the ChatGPT attribution from Studies 1 and 2 (AI-authored condition), "Letter prepared by OpenAI's ChatGPT Artificial Intelligence language model and edited by Sandy Lewis" (AI-authored/Human-edited), or "Letter prepared by Sandy Lewis and edited by OpenAI's ChatGPT Artificial Intelligence language model" (Human-authored/AI-edited). We measured moral disgust, positive word of mouth, and loyalty as in Studies 1 and 2.

5.2. Results and discussion

An ANOVA revealed that PWOM differed across conditions ($F(3, 396) = 15.57$, $p < 0.001$, $\eta^2 = 0.11$; see Fig. 4). Follow-up contrasts with Tukey HSD corrections revealed that relative to the Human-authored condition ($M = 5.34$, $SD = 1.16$, participants in both the AI ($M = 4.08$, $SD = 1.57$, $t = -6.30$, $p < 0.001$, $\eta^2 = 0.09$) and AI-authored/Human-edited ($M = 4.43$, $SD = 1.58$, $t = -4.55$, $p < 0.001$, $\eta^2 = 0.05$) conditions reported lower PWOM. Importantly, PWOM in the Human-authored/AI-edited condition ($M = 4.97$, $SD = 1.32$) was greater than in the AI-authored condition ($t = 4.50$, $p < 0.001$, $\eta^2 = 0.05$). It was also greater than in the AI-authored/Human-edited condition ($t = 2.70$, $p = 0.036$, $\eta^2 = 0.02$). The two Human-authored and two AI-authored conditions did not differ from each other ($p = 0.269$ and $p = 0.283$ respectively; see Table WA4 for full results of all condition

[2] In Study 2, we also measured moral anger and perceived moral wrongness. Including these variables in our model did not change our results (see data files).

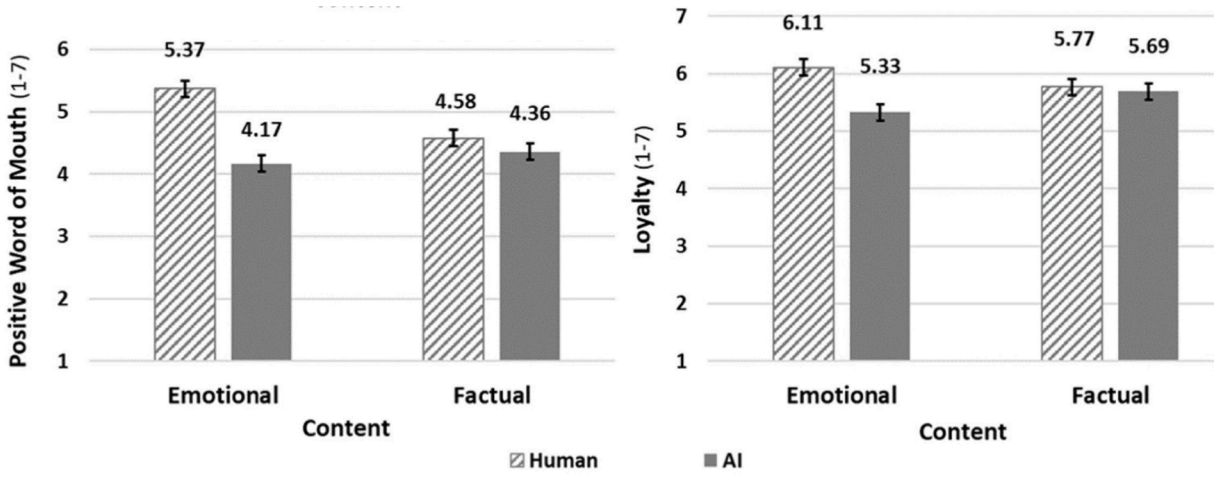


Fig. 2. Authorship Effects on Positive Word of Mouth and Customer Loyalty Are Attenuated when the Content Is Less Emotional (Study 2).

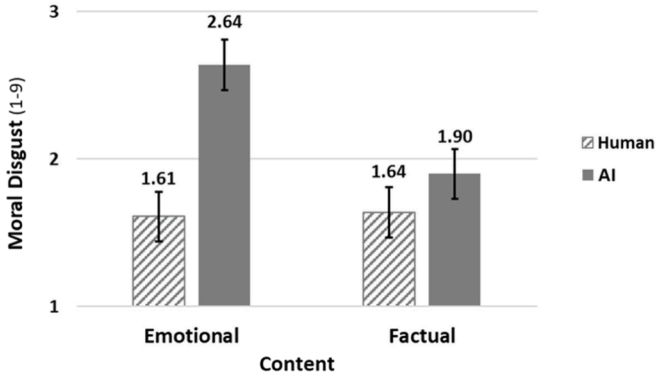


Fig. 3. Authorship Effects on Moral Disgust Are Attenuated when the Content Is Less Emotional (Study 2).

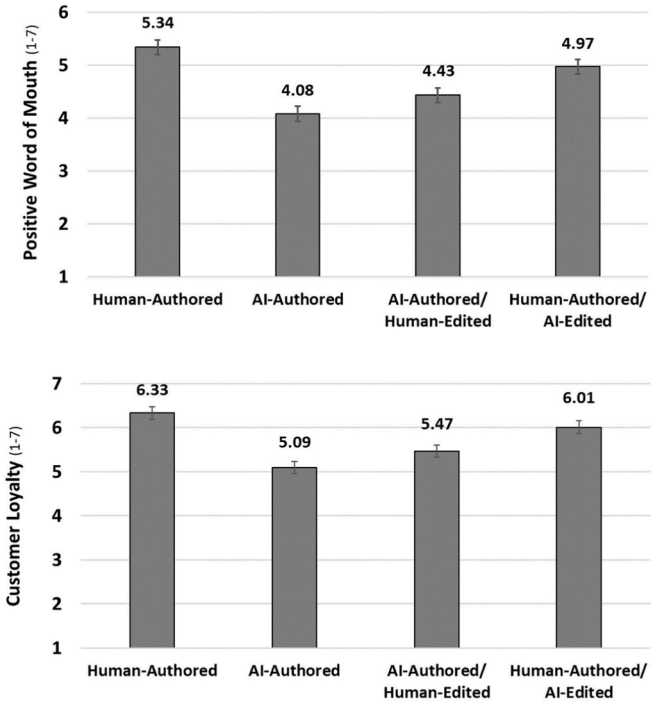


Fig. 4. Editing (vs. Authoring) by AI Increases PWOM and Customer Loyalty (Study 3).

comparisons).

Loyalty also differed across conditions ($F(3, 396) = 16.41$, $p < 0.001$, $\eta^2 = 0.11$). Relative to the Human-authored condition ($M = 6.33$, $SD = 1.07$), participants in both the AI-authored ($M = 5.09$, $SD = 1.66$, $t = -6.53$, $p < 0.001$, $\eta^2 = 0.10$) and AI-authored/Human-edited ($M = 5.47$, $SD = 1.44$, $t = -4.53$, $p < 0.001$, $\eta^2 = 0.05$) conditions reported lower loyalty. Importantly, loyalty in the Human-Authored/AI-edited condition ($M = 6.01$, $SD = 1.17$) was significantly greater than in the AI-authored condition ($t = 4.84$, $p < 0.001$, $\eta^2 = 0.05$). It was also greater than in the AI-authored/Human-edited condition ($t = 2.79$, $p = 0.028$, $\eta^2 = 0.02$). The two Human-authored and two AI-authored conditions did not differ from each other ($p = 0.345$ and $p = 0.189$ respectively).

Moral disgust also differed across conditions ($F(3, 396) = 13.34$, $p < 0.001$, $\eta^2 = 0.09$; see Fig. 5). Relative to the Human-authored condition ($M = 1.61$, $SD = 1.18$), participants in both the AI-authored ($M = 3.23$, $SD = 2.39$, $t = 5.79$, $p < 0.001$, $\eta^2 = 0.08$) and AI-authored/Human-edited ($M = 3.00$, $SD = 2.15$, $t = 2.96$, $p < 0.001$, $\eta^2 = 0.06$) conditions reported more moral disgust. Importantly, moral disgust in the Human-authored/AI-edited condition ($M = 2.44$, $SD = 1.93$) was significantly less than in the AI-authored condition ($t = -2.82$, $p = 0.025$, $\eta^2 = 0.02$). The Human-authored condition also elicited lower moral disgust than the Human-authored/AI-edited condition ($t = -2.96$, $p = 0.017$, $\eta^2 = 0.02$). Disgust in the two AI-authored conditions did not differ ($p = 0.843$). Disgust in the two edited conditions (Human-authored/AI-edited and AI-authored/Human-edited) also did not differ ($t = -2.00$, $p = 0.182$).

Replicating our prior results, a mediation analysis with the AI-authored condition as the baseline (Hayes, 2018; PROCESS model 4) revealed a significant indirect effect of authorship through moral disgust on PWOM for the Human-authored (vs. AI-authored) condition ($B_{Human} = 0.69$, $SE = 0.12$, $CI_{95\%}$ [0.46, 0.95]. Importantly, supporting hypothesis 5a, the indirect effect of authorship on PWOM was also significant for the Human-authored/AI-edited (vs. AI-authored) condition ($B_{Human\text{-}authored/AI\text{-}edited} = 0.34$, $SE = 0.14$, $CI_{95\%}$ [0.08, 0.61]). The indirect effect for the AI-authored/Human-edited (vs. AI-authored) condition was not significant ($B_{AI\text{-}authored/Human\text{-}edited} = 0.10$, $SE = 0.14$, $CI_{95\%}$ [−0.17, 0.37]; Table WA5).

Again, using the AI-authored condition as the baseline (Hayes, 2018; PROCESS model 4), we obtained similar indirect effects on loyalty for the Human-authored (vs. AI-authored) condition ($B_{Human} = 0.71$, $SE = 0.12$, $CI_{95\%}$ [0.48, 0.95]. Importantly, supporting hypothesis 5b, the indirect effect of authorship on PWOM was significant for the Human-authored/AI-edited (vs. AI-authored) condition ($B_{Human\text{-}authored/AI\text{-}}$

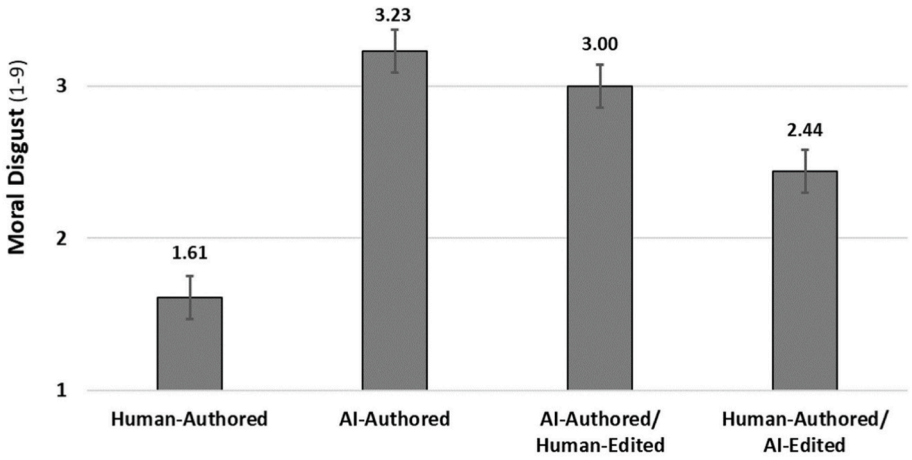


Fig. 5. Editing (vs. Authoring) by AI Decreases Moral Disgust (Study 3).

$_{edited}$ = 0.34, *SE* = 0.14, $CI_{95\%}$ [0.08, 0.62]). The indirect effect for the AI-authored/Human-edited (vs. AI-authored) condition was not significant ($B_{AI\text{-}authored/\ Human\text{-}edited}$ = 0.10, *SE* = .14, $CI_{95\%}$ [−0.17, 0.37]; Table WA5).

In Study 3, we replicated the AI-authorship effect, this time with a negative emotion: sympathy. Further, supporting hypotheses 5a-b, we found that the negative impacts of AI authorship attenuate when AI is used to edit, rather than write, the communication.

6. Study 4: The AI-authorship effect attenuates when the AI signs the communication

We have argued that when an AI signs an emotional communication directly, the message will elicit less moral disgust than if it was generated by an AI yet signed by a human because the AI will be seen as more autonomous. In this case, we predicted attenuation of the AI-authorship effect. We tested these hypotheses (6a-b) in Study 4 using the emotion of pride.

6.1. Method

Study 4 (preregistered: https://aspredicted.org/M7V_3P6) employed a single-factor, three condition design (Author: Human; AI signed by Human; AI signed) with 300 U.S. and U.K. Prolific participants. Participants imagined they were a long-time customer of Summit Apparel, a small high-performance apparel company, and that they had received an email (Web Appendix):

> *We are overflowing with pride and gratitude as we share some exciting news with you. Summit Apparel has been honored with the prestigious Innovation Award from the Outdoor Apparel Trade Association! … We couldn't be more proud of our incredible team… I know that their ingenuity and tireless pursuit of perfection are the driving forces behind the revolutionary technologies we've pioneered. With pride and gratitude.*

The email was signed either by "Sandy Robbins, Vice President of Summit Apparel" (human), by "The Summit Apparel Artificial Intelligence ChatBot" (AI signed) or by Sandy Robbins with the ChatGPT attribution from Study 1 (AI signed by human). We measured moral disgust, PWOM, and loyalty as previously.[3]

To confirm that an AI chatbot signing an email directly (vs. an AI chatbot preparing an email for a human to sign) is perceived as more autonomous, we conducted a pretest with 169 Prolific participants. We used the main study stimuli and measured perceived autonomy with two 7-point items ("The AI chatbot seems to… "control its own actions" and "be able to decide on its own whether to communicate with me"; *r* = 0.69, *p* < 0.001). The chatbot signing directly was perceived as more autonomous (*M* = 3.63, *SD* = 1.57) than the chatbot preparing an email for a human to sign (*M* = 3.10, *SD* = 1.41; *F*(1, 167) = 5.23, *p* = 0.023, η^2 = 0.03).

6.2. Results and discussion

An ANOVA revealed that PWOM differed across conditions (*F*(2, 297) = 19.60, *p* < 0.001, η^2 = 0.12; see Fig. 6). Follow-up contrasts with Tukey HSD corrections revealed that PWOM was lower when the email was written by AI but signed by a human (*M* = 3.84, *SD* = 1.54) than when either the human signed it (*M* = 5.13, *SD* = 1.43; *t* = −6.22, *p* < 0.001, η^2 = 0.12), or when the AI chatbot signed it (*M* = 4.34, *SD* = 1.43; *t* = −2.43, *p* = 0.016, η^2 = 0.02). Similarly, loyalty differed across conditions (*F*(2, 297) = 13.70, *p* < 0.001, η^2 = 0.08). Loyalty was lower when the email was written by AI but signed by a human (*M* = 5.02, *SD* = 1.52) than when either the human signed it (*M* = 6.04, *SD* = 1.28; *t* = −5.20, *p* < 0.001, η^2 = 0.08), or when the AI chatbot signed it (*M* = 5.45, *SD* = 1.32; *t* = −2.22, *p* = 0.069, η^2 = 0.02). PWOM (*t* = −3.81, *p* < 0.001, η^2 = 0.05) and loyalty (*t* = −3.01, *p* = 0.008, η^2 = 0.03) were also lower in the AI-signed than in the human-signed conditions.

Moral disgust also differed across conditions (*F*(2, 297) = 27.65, *p* < 0.001, η^2 = 0.16). Moral disgust was higher when the email was written by AI but signed by a human (*M* = 3.30, *SD* = 2.36) than when either the human signed it (*M* = 1.37, *SD* = 0.94; *t* = 7.42, *p* < 0.001, η^2 = 0.16), or when the AI chatbot signed it (*M* = 2.26, *SD* = 1.89; *t* = 4.04, *p* < 0.001, η^2 = 0.05). Disgust was also higher in the AI-signed than in the human-signed condition (*t* = 3.42, *p* = 0.002, η^2 = 0.04). Mediation analysis (Hayes, 2018; PROCESS model 4) confirmed that, compared with the AI-authored and human-signed condition, the relative indirect effects of both human signing (B = .83, *SE* = 0.13, $CI_{95\%}$ = [0.58, 1.10]) and AI signing (B = 0.45, *SE* = 0.13, $CI_{95\%}$ = [0.19, 0.73]) on PWOM through moral disgust were significant. Similarly, compared with the AI-authored human-signed condition, the relative indirect effects of both human signing (B = 0.78, *SE* = 0.13, $CI_{95\%}$ = [0.55, 1.05]) and AI signing (B = 0.42, *SE* = 0.13, $CI_{95\%}$ = [0.18, 0.69]) on loyalty through moral disgust were significant. Thus, hypotheses 6a-b were supported.

Using a different positive emotion, pride, the results of Study 4 reveal an important boundary condition with substantial implications for managers. Specifically, when an AI chatbot signs an emotional marketing communication directly, rather than having the AI-generated communication signed by a human, moral disgust decreases, resulting in increased PWOM and loyalty.

[3] We also included a measure of intention to join a loyalty program on an exploratory basis (see data files).

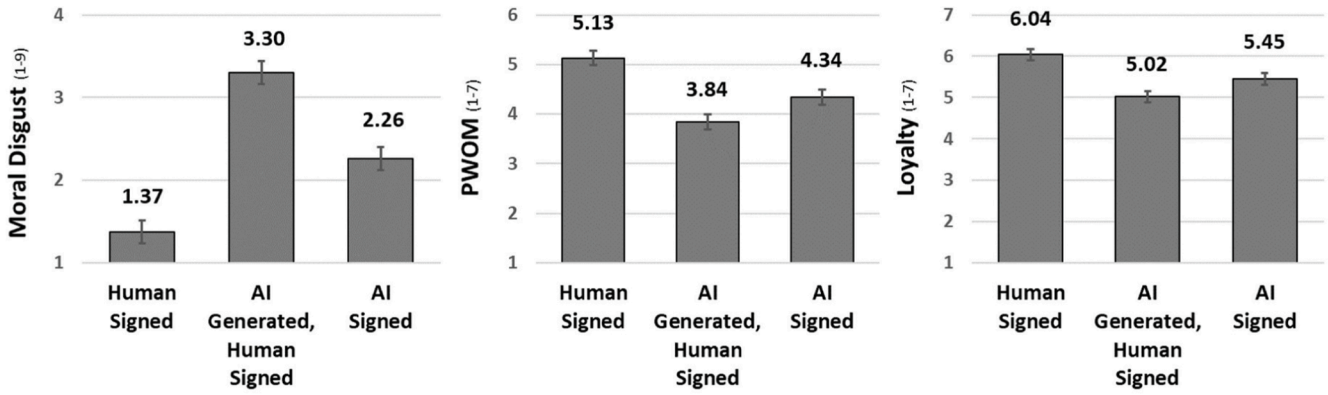


Fig. 6. The AI-Authorship Effect Is Attenuated When the AI Signs the Communication Directly (Study 4).

7. Study 5: Serial mediation by perceived authenticity and moral disgust

We have argued that consumers will perceive emotional communications written by an AI (versus a human) to be inauthentic, and therefore that the AI-authorship effect is serially mediated by perceived inauthenticity and moral disgust (hypotheses 3a-c). In Study 5, we use a different positive emotion, gratitude, to test these hypotheses.

7.1. Method

Study 5 (preregistered: https://aspredicted.org/VLR_11G) employed a single-factor design (Author: Human vs. AI) with 200 U.S. and U.K. Prolific participants, using the same company, Avery's Organics, from Study 2. They imagined receiving an email from the new Sales Director, Pat Robbins:

> *As I sit down to write this letter, my heart overflows with gratitude for each and every one of you. You are the reason we exist as a company and the reason we strive to make the best possible personal care products on the market…So, from the bottom of my heart, I want to say thank you… We could not do this without you. With warmth and appreciation.*

We manipulated authorship as in Study 1. We measured perceived authenticity with four 7-point bipolar items (Napoli et al., 2014): "Pat Robbins was…" "Not true to their own values/True to their own values"; "Inauthentic/Authentic"; "Lacking in principles/Highly principled"; "Compromising their values/Not compromising their values" ($\alpha = 0.93$). We measured moral disgust ($\alpha = 0.91$), PWOM ($\alpha = 0.97$) and loyalty ($\alpha = 0.95$) as previously.

7.2. Results and discussion

Replicating our prior results, PWOM was higher when the email was written by the human ($M = 5.12$, $SD = 1.35$) than the AI ($M = 4.02$, $SD = 1.57$; $F(1, 198) = 28.52$, $p < 0.001$, $\eta^2 = 0.13$). Loyalty followed a similar pattern ($M_{Human} = 6.02$, $SD = 1.20$ vs. $M_{AI} = 5.04$, $SD = 1.52$; $F(1, 198) = 25.43$, $p < 0.001$, $\eta^2 = 0.11$). Further, moral disgust was lower for the human ($M = 1.84$, $SD = 1.43$) than for the AI ($M = 3.44$, $SD = 1.82$; $F(1, 198) = 48.09$, $p < 0.001$, $\eta^2 = 0.20$). Importantly, supporting hypothesis 3a, perceived authenticity was higher for the human ($M = 5.38$, $SD = 1.37$) than for the AI ($M = 3.99$, $SD = 1.61$; $F(1, 198) = 43.08$, $p < 0.001$, $\eta^2 = 0.18$).

Results of serial mediation analyses (Hayes, 2018; PROCESS Model 6) revealed that the indirect effect of authorship through perceived authenticity and moral disgust was significant ($B = -0.17$, $SE = 0.05$, $CI_{95\%} = [-0.28, -0.07]$). Similarly, the indirect effect on loyalty was also significant ($B = -0.33$, $SE = 0.09$, $CI_{95\%} = [-0.52, -0.18]$). Thus, hypotheses 3a-c were supported.

In sum, Study 5 verified the key role that perceived authenticity plays in explaining the AI-authorship effect. Specifically, authenticity requires a perception that an entity is communicating a true and honest internal state. An AI does not have an internal state; therefore, when a human uses an AI to generate an emotional communication, it is not seen as authentic, leading to moral disgust and reducing PWOM and loyalty.

8. Study 6: The AI-authorship effect reverses when the sender reuses emotional content

Authenticity theory suggests there are often situations when humans might be seen as inauthentic communicators. One of these should be when they copy and reuse emotional content originally created for another purpose. In this case, a human communicator would be seen as inauthentic due to a deliberate reluctance to express true internal values. However, an AI is not driven by personal motivations, and thus when AI reuses or repurposes existing content, it should incur a smaller authenticity penalty, reversing the AI-authorship effect (hypotheses 7a-b). We test these moderated serial mediation hypotheses in Study 6.

8.1. Method

Study 6 (preregistered: https://aspredicted.org/WD7_LZG) employed a 2 (Author: Signed by Human vs. Signed by AI) × 2 (Content: Original vs. Copied) factor design with 600 U.S. Prolific participants (222 male, 373 female, 5 other; $M_{Age} = 42.96$, $SD = 13.11$), using the same company and email from Study 5.[4] Participants read that the email was signed by either Sandy Robbins, Sales Director (human condition) or Sandy, the Avery Organics Artificial Intelligence Chatbot (AI condition). They also read, "After you read the email, you realize the company has replaced the previous sales director (chatbot), who (which) was named Pat Collins (Pat)." In the copied conditions, participants further read, "You realize that the email is virtually identical to the email the previous sales director (chatbot) sent last year." They then read the We measured perceived authenticity ($\alpha = 0.94$), moral disgust ($\alpha = 0.93$), PWOM ($\alpha = 0.96$), and loyalty ($r = 0.89$), as in Study 5, as well as content originality as a manipulation check ("How original was the email content?" 1 = "Copied from somewhere else"; 7 = "Original [generated by the sender]").

8.2. Results and discussion

The copied content was less original ($M = 1.78$, $SD = 1.51$) than the non-copied content ($M = 5.10$, $SD = 1.74$; $F(1, 596) = 639.57$; $p < 0.001$, $\eta^2 = 0.52$). The manipulation was successful.

A two-way ANOVA on PWOM revealed main effects of authorship (F

[4] For this study, we simplified the company name to Avery Organics.

(1, 596) = 15.33, $p < 0.001$, $\eta^2 = 0.03$) and content ($F(1, 596) = 105.41$, $p < 0.001$, $\eta^2 = 0.15$). Importantly, the interaction was also significant ($F(1, 596) = 39.60$, $p < 0.001$, $\eta^2 = 0.06$; Fig. 7). Replicating our prior results, when the content was original, AI authorship reduced PWOM ($M_{Human} = 5.35$, $SD = 1.31$ vs. $M_{AI} = 4.16$, $SD = 1.62$; $F(1, 596) = 51.93$, $p < 0.001$, $\eta^2 = 0.08$). However, when the content was copied, the effect was reversed. Instead, AI authorship increased PWOM compared with the human author ($M_{Human} = 3.41$, $SD = 1.28$ vs. $M_{AI} = 3.69$, $SD = 1.51$; $F(1, 596) = 2.84$, $p = 0.093$, $\eta^2 = 0.01$).[5] Similarly, for loyalty, we also found main effects of authorship ($F(1, 596) = 4.31$, $p = 0.038$, $\eta^2 = 0.01$) and content ($F(1, 596) = 41.40$, $p < 0.001$, $\eta^2 = 0.07$), and a significant interaction ($F(1, 596) = 38.03$, $p < 0.001$, $\eta^2 = 0.06$). Again, when the content was original, AI authorship reduced loyalty ($M_{Human} = 6.11$, $SD = 1.26$ vs. $M_{AI} = 5.06$, $SD = 1.63$; $F(1, 596) = 33.87$, $p < 0.001$, $\eta^2 = 0.05$). However, when the content was copied, the effect was reversed. Instead, AI authorship increased loyalty compared with the human author ($M_{Human} = 4.50$, $SD = 1.60$ vs. $M_{AI} = 5.02$, $SD = 1.70$; $F(1, 596) = 8.39$, $p = 0.004$, $\eta^2 = 0.01$).

We found similar results for both perceived authenticity and moral disgust. For authenticity, we found main effects of authorship ($F(1, 596) = 19.04$, $p < 0.001$, $\eta^2 = 0.03$) and content ($F(1, 596) = 205.44$, $p < 0.001$, $\eta^2 = 0.26$) and a significant interaction ($F(1, 596) = 52.67$, $p < 0.001$, $\eta^2 = 0.08$). When the content was original, authenticity was higher in the human ($M = 5.63$, $SD = 1.23$) than in the AI ($M = 4.28$, $SD = 1.50$) condition ($F(1, 596) = 67.29$, $p < 0.001$, $\eta^2 = 0.10$). However, when the content was copied, the effect was reversed ($M_{Human} = 3.12$, $SD = 1.40$ vs. $M_{AI} = 3.46$, $SD = 1.54$; $F(1, 596) = 4.20$, $p = 0.041$, $\eta^2 = 0.01$). Similarly, for moral disgust, we found a marginally significant main effect of authorship ($F(1, 596) = 3.06$, $p = 0.081$, $\eta^2 = 0.01$) and a main effect of content ($F(1, 596) = 65.92$, $p < 0.001$, $\eta^2 = 0.10$), along with a significant interaction ($F(1, 596) = 40.43$, $p < 0.001$, $\eta^2 = 0.06$). When the content was original, disgust was lower in the human ($M = 1.46$, $SD = 1.06$) than in the AI ($M = 2.75$, $SD = 2.16$) condition ($F(1, 596) = 32.76$, $p < 0.001$, $\eta^2 = 0.05$). However, when the content was copied, the effect was reversed ($M_{Human} = 3.77$, $SD = 2.15$ vs. $M_{AI} = 3.03$, $SD = 2.22$; $F(1, 596) = 10.66$, $p < 0.001$, $\eta^2 = 0.02$).

Results of a moderated serial mediation analysis (Hayes, 2018; PROCESS Model 83; Fig. 8) for PWOM revealed a significant index of moderated mediation (B = 0.15, $SE = .04$, $CI_{95\%} = [0.08, 0.24]$). When the content was original, the effect of Human (=0) vs. AI (=1) authorship on PWOM through perceived authenticity and moral disgust was negative (B = −0.12, $SE = 0.03$, $CI_{95\%} = [-0.19, -0.07]$. However, when the content was copied, the serially mediated effect was reversed (B = 0.03, $SE = 0.02$, $CI_{95\%} = [0.001, 0.07]$). We obtained similar results for loyalty (B = 0.43, $SE = 0.08$, $CI_{95\%} = [0.29, 0.58]$). When the content was original, the effect of Human (vs. AI) authorship through perceived authenticity and moral disgust was negative (B = −0.34, $SE = 0.05$, $CI_{95\%} = [-0.45, -0.24]$. However, when the content was copied, the serially mediated effect was reversed (B = 0.08, $SE = 0.04$, $CI_{95\%} = [0.001, 0.17]$). Therefore, hypotheses 7a-b were supported.

In sum, when humans copy and reuse emotional marketing communications, they are perceived as inauthentic, eliciting moral disgust and reducing PWOM and loyalty. In contrast, an AI communicator is less likely to face such an authenticity penalty, leading to a reversal of the AI-authorship effect when emotional marketing content is reused. This finding underscores the critical role of perceived authenticity in shaping consumer responses to marketing messages. It also suggests opportunities for marketers to leverage AI to enhance consumer outcomes in situations where reusing emotional content could be seen as inauthentic coming from a human.

[5] When the nine participants who failed the attention check were excluded, this difference reached significance (see Web Appendix).

9. General discussion

With this research, we document for the first time an authorship effect from the use of generative AI in marketing communications. Across six preregistered studies using both hypothetical and actual behavioral outcome measures, we have demonstrated that PWOM and loyalty are reduced when consumers believe that emotional marketing communications have been generated by an AI (vs. a human). We have shown that AI-authored emotional communications are seen as morally inauthentic (Studies 5–6), leading to feelings of moral disgust and explaining these effects. We have demonstrated these effects using both individual (Study 1) and mass communications from different organizational employees and different businesses expressing different emotions (inspiration, happiness, gratitude, sympathy, and pride). Drawing from authenticity theory, we document important managerially-actionable moderators. We have shown that this AI-authorship effect is evident only when an AI generates emotional (but not factual) communications (Study 2); that the effect is diminished when an AI simply edits (vs. authors) a marketing communication (Study 3); and when the AI signs the communication directly (Study 4). Finally, we show that when emotional marketing communications are copied and reused, the AI-authorship effect reverses, with the AI being seen as more authentic than the human, leading to more positive outcomes for marketers (Study 6).

9.1. Implications for theory

This research provides a first step toward answering calls for understanding how the disclosure of information about the use of generative AI will change consumer behavior (Hermann & Puntoni, 2024), contributing to the nascent research on generative AI in marketing. In doing so, we also contribute to the growing literature documenting the vital impact of authenticity on consumers' responses to marketing communications (Beverland et al., 2008; Cascio Rizzo et al., 2023; Zhang & Patrick, 2021). Authenticity refers to the congruence between an entity's internal state, or values, and its external communications (Beverland et al., 2008; Newman, 2019; Zhang & Patrick, 2021). An AI is a machine and has no internal emotional states. In accord, we show that the AI-authorship effect is serially mediated by perceived authenticity and moral disgust. Further, factual content is less likely than emotional content to purport to communicate an internal state. Accordingly, we document that the AI-authorship effect is attenuated when the communication content is factual and when an AI only edits, rather than authors, emotional content. Finally, copying and reusing emotional content that another entity has written for a totally unrelated situation should also be inauthentic because the individual is not actually feeling the expressed emotional state. We document that in this situation, human authors suffer an inauthenticity penalty, whereas AI authors are less likely to do so, resulting in a reversal of the AI-authorship effect.

We also add to the limited literature on moral disgust in marketing (Amar et al., 2018; Grappi et al., 2013) by illustrating for the first time that consumers feel moral disgust from emotional marketing communications generated by an AI. Moral disgust stems from violations of moral principles (Chapman & Anderson, 2013). If a marketer's communication is inauthentic (not consistent with their internal state), they are misrepresenting themselves and therefore being dishonest to their customers. In accord, we document that inauthenticity in marketing communications leads to moral disgust. We also show that if an AI can be seen as more of an autonomous social entity, its attempts at emotional communications elicit less moral disgust. In accord, we show that the AI-authorship effect attenuates when the AI signs an emotional communication directly (Study 4). Finally, we also contribute to the word of mouth and loyalty literatures by showing that, consistent with disgust as an emotion that elicits avoidance responses, moral disgust from marketers' communications reduces positive word of mouth and customer loyalty.

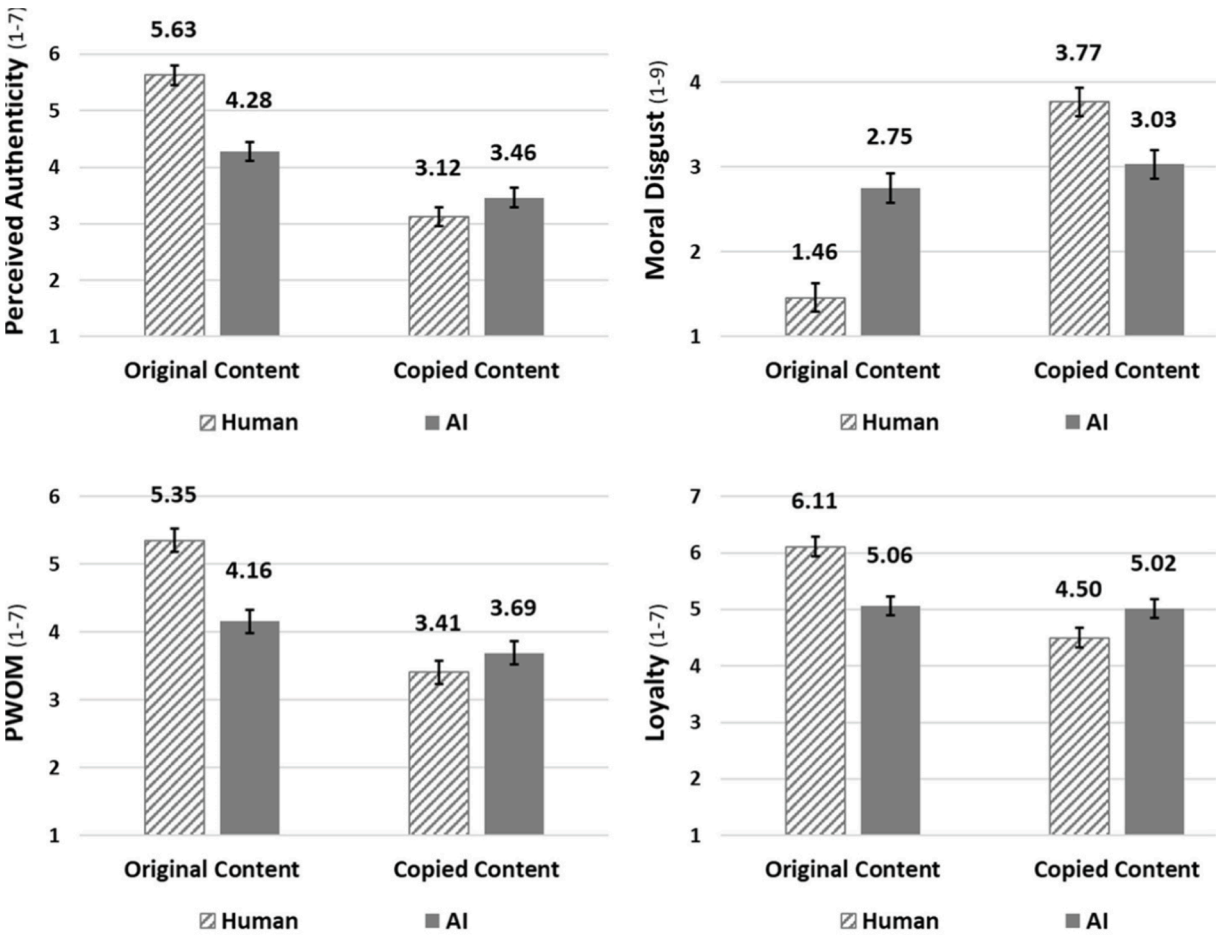
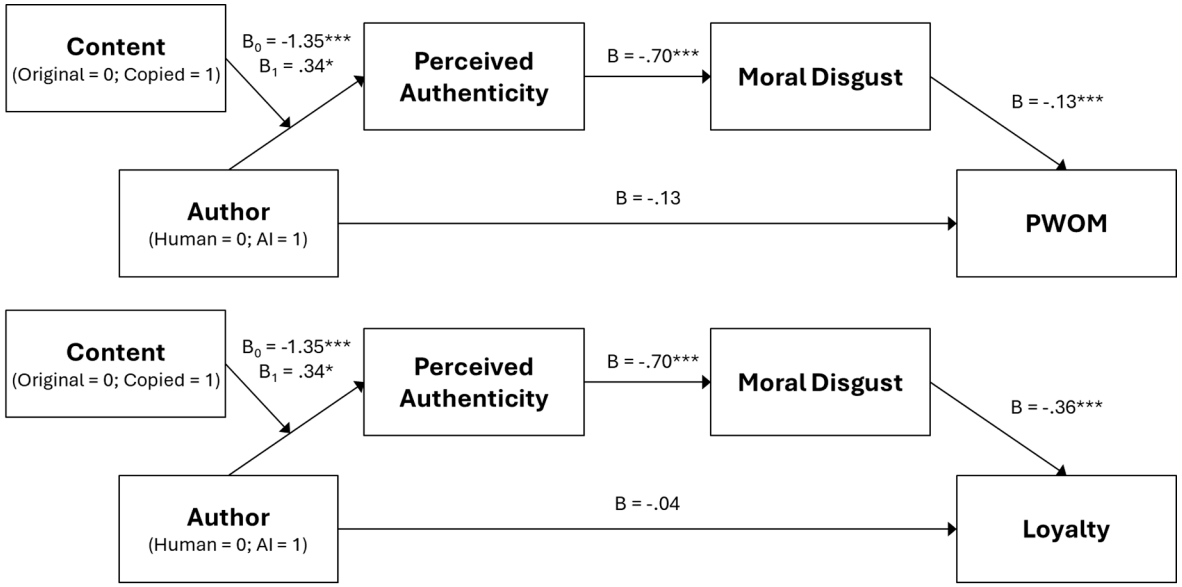


Fig. 7. The AI-Authorship Effect Reverses for Copied and Reused Communications (Study 6).



Note. Values indicate unstandardized regression coefficients. *$p < .05$. ***$p < .001$.

Fig. 8. Moderated Serial Mediation Results (Study 6).

9.2. Practical implications

An immediate practical implication is that companies must consider whether and how to disclose AI-authored communications. Given increasing regulations mandating AI disclosure, such as the European Union AI Act (European Commission, 2024) and other recent and pending legislation in the U.S. (Levi et al., 2024), as well as demands for transparency amid authenticity concerns from consumers (Yahoo, 2024), these are pressing questions. This research suggests that disclosing AI authorship for factual content may not negatively impact consumer responses. To be perceived as authentic, a communication must be consistent with a communicator's "internal state." Factual communications are not intended to convey emotions, so AI authorship creates less inconsistency between internal states and external expressions (i.e., less inauthenticity) than with emotional communications. However, companies often use emotional content to strengthen customer engagement and brand attachment. Our findings suggest this effort will backfire when customers know or believe a communication is AI-written. They will view such communications as inauthentic, responding with moral disgust and penalizing the firm with reduced positive word of mouth and loyalty. Further, this research highlights the risk to firms of not disclosing AI use in content creation if it is later exposed or seriously suspected by consumers, such as through leaks, retroactive legal disclosure requirements, or new methods to detect AI communication patterns.

However, we also show that using an AI to edit emotional communications has less of an impact than using an AI to prepare them. Editing is a process that is less likely to impact the authenticity of the original communication, providing marketers a way to leverage this technology without eliciting a backlash from consumers. Further, we show that when marketers are obligated to reveal that their emotional communications were generated by AI, they will be better off having an AI sign such communications directly. At a minimum, an AI will be less likely to elicit negative outcomes than the same email signed by a human but generated by AI. And when communications are reused for different purposes, as they often are, an AI will be seen as more authentic than a human, thus eliciting more positive outcomes. Our findings also suggest implications for policymakers and technology firms as they determine the level of disclosure required when producing content using AI technology (Fitch, 2024).

More broadly, our findings suggest that companies need to strive for authenticity in their communications. For example, in Study 6, our findings that inauthentic human communications, such as those copied from someone else, elicit moral disgust and reduce PWOM and loyalty can be applied in non-AI contexts. Service providers, such as an airline, might consider whether live emotional announcements, such as those expressing happiness at welcoming customers, might be preferable to commonly-used prerecorded announcements.

As the use of generative AI becomes increasingly widespread, consumers are likely to become ever more skeptical of the human origin of marketers' communications, making the AI-authorship effect an increasing concern for marketers. To investigate this possibility, we conducted a seventh preregistered experiment (Study WA1; see Web Appendix) in which we manipulated both AI authorship as in Study 6, as well as participants' beliefs that most marketers' messages are written by AI (vs. a human). We found that when participants believed most marketers are using AI (vs. humans) to write their communications, participants felt disgust and responded negatively regardless of whether an emotional email was signed by an AI or a human. When they believed the opposite, we replicated our typical pattern. These findings suggest that as AI authorship becomes more normative, there will be less of a penalty for using AI instead of a human to write emotional content. Further, marketers may increasingly benefit from highlighting when their content is human-authored, especially if it is original and emotional. Indeed, some marketers are already disavowing AI involvement in some of their communications. For example, in its "The Code" ad campaign, Dove promises never to use AI to create or change pictures of women (Dove, 2023) and Etsy's ad campaign for the 2024 Olympics highlights the humans making and sourcing its products (Pasquarelli, 2024).

9.3. Limitations and future research directions

Our work has limitations. For example, although one of our studies used actual word of mouth behaviors, the remainder used hypothetical scenarios. We also recruited participants from online platforms. Despite the popularity of such platforms (Peer et al., 2022), it is preferable to vary participant sources. Further, authenticity perceptions differ across cultures (Sedikides & Schlegel, 2024), and our study participants resided in the U.S. or U.K. Future investigations could address these limitations.

Our work opens empirical avenues for future research. Consumers who are more technologically innovative (Bruner & Kumar, 2007) might be more forgiving of using an AI to express one's internal states. Further, we intentionally focused on marketing communications due to the importance of the AI-authorship effect for business. However, one might expect to find the AI-authorship effect in consumer-to-consumer communications (Hass et al., 2024) as well. For example, if a best man needed to write an emotional toast to the happy couple (O'Gieblyn, 2023), our findings suggest that he would be well-served not to admit AI assistance. Our framework also suggests that consumers who are higher in advertising skepticism (Obermiller et al., 2005) might be less influenced by the AI-authorship effect, as they may perceive all marketing communications as inauthentic. Anthropomorphizing an AI (Garvey et al., 2023) might also increase perceptions of authenticity in AI-generated marketing communications, thereby attenuating the AI-authorship effect. It is also possible that human curation of AI output, beyond simply editing, might also attenuate the effect. Our findings may also be impacted by the nature of the AI technology used for the communication (e.g., virtual influencers, virtual agents, robots, etc.), suggesting other avenues for further research. The use of other generative AI technologies beyond large language models, such as image, video and audio communications, could also be tested, likely with similar outcomes.

The relationship between the consumer and the firm might matter, too. Prior research has established that the relationship strength between a customer and a firm increases customers' perceived betrayal after a service failure (Grégoire et al., 2009). It is possible that such a strong relationship or feelings of ownership of a message recipient for a brand (Kirk et al., 2018) might also augment the AI-authorship effect. Our findings also suggest that the AI-authorship effect would not be evidenced if the AI-authored communication refers to imagining how a *human* might express an emotion (rather than expressing the emotion itself). An example of this is the AI-generated "Jesus Gets Us" advertisements appearing during the 2023 Superbowl (Nudd, 2023), in which the AI imagines how Jesus would express love.

CRediT authorship contribution statement

Colleen P. Kirk: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Julian Givi:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data for this research is available in the authors' OSF project repository: https://osf.io/ra9hz/?view_only=e776c99ca0d44987850e851c9c4ca5f4

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jbusres.2024.114984.

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